

Perceptual Encryption of JPEG Compressed Images Using DCT Coefficients and Splitting of DC Coefficients into Bitplanes

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Abstract- Perceptual Encryption is required in many of the multimedia content preview application. There is always a need of an efficient technique that reduces the overhead of encryption in real-time. In this paper we take advantage of the energy present in DCT block and have studied the effects of encrypting the selected DCT coefficient. Also the DC coefficient is further divided into bitplanes and an investigation is carried out to study the resulting depth of perceptual encryption. To measure the level of perceptual degradation, objective image quality assessment metrics such as SSIM, MS-SSIM, VIF, VIFP, VSNR, IFC and UQI are used. The presented results can help one chose the desired level of perceptual encryption according to the application with very little overhead.

I. INTRODUCTION

Past decade has witnessed extensive amount of transmission of visual data like images and videos via internet. Because this kind of data is sometimes sensitive in nature so the issues related to the security and privacy of these multimedia contents needs serious consideration. A huge amount of work has been done in this area, after the emergence of popular image and video compression standards (i.e. JPEG and MPEG). This comprehensive study on multimedia content encryption techniques can be found in [1].

This growth in the multimedia technologies has resulted in the development of many commercial applications like video on demand (VoD), pay-TV, pay-per-view etc. [2]. Few of the applications regarding the perceptual encryption of images are mentioned in [6]. This kind of multimedia application requires multimedia content preview, where in the multimedia content (usually a video) is degraded to some poor quality so that it can be preview free of charge and if the consumer wants to watch the high quality video then she/he has to pay for watching it. Perceptual encryption deals with the degraded quality of the multimedia content according to the desired requirements. One of the applications of the perceptual encryption is the multimedia content preview. A lot of work has been done in this area [3]-[9]. A comprehensive survey on the related work is presented in [10].

In this paper an investigation has been done on the effect of encrypting DCT coefficients and along the lines of [11]. However, in the work in [11] is further refined by dividing DC Coefficient in Discrete Cosine Transform block, in to bitplanes and these bitplanes are encrypted and also the quality of the encrypted image in measured with several

metrics. For the purpose of encryption we used the simplest of the encryption technique that is scrambling. More specifically a chaotic scrambler, that is described briefly in the later section. However, the scrambler chosen has an optimal property of spread and dispersion [12]. And for the purpose of measuring the level of perceptual degradation introduced in the images, objective image quality metrics such as SSIM [13], MS-SSIM [14], VIF [15], VIFP [15], UQI [16] and IFC [17] are used. Also PSNR [18] values have been measured in order to make one able to compare results with the exiting schemes.

The remainder of this paper is organized as follows: related work is presented in Section 2, Section 3 focuses on the methodology of the proposed scheme and Section 4 and 5 emphasizes on scrambler used to scramble the multimedia data and the method of performance evaluation respectively. An analysis has been done and results are presented in section 7. And in last some conclusions have been drawn in Section 8.

II. RELATED WORK

In [19], a selective encryption or soft encryption scheme for uncompressed images is proposed, in which the image is decomposed in to 8 bitplanes (i.e. $i_7, i_6, i_5, i_4, i_3, i_2, i_1, i_0$). One by one, starting from the Least Significant Bitplane up to Most significant Bitplane are encrypted. Mean Square Error (MSE) and Peak Signal to Noise Ratio (PSNR) is used as a metric to measure the distortion introduced in the encrypted images. Linear relationship has been shown between the PSNR values and the number of encrypted bitplanes. Also a method to encrypt JPEG compressed images is proposed in [19], where the appended bits¹ (specify the sign and magnitude of non-zero coefficients) of selected AC coefficients are encrypted. The metrics used to measure the encrypted image quality are only MSE and PSNR and classical encryption algorithm are used like DES, triple DES and IDEA. In [20], another selective encryption approach is proposed to secure uncompressed image data transmission in mobile environment. In the proposed scheme the image is divided in to 8 bitplanes and starting from the Most Significant Bitplane each bitplane is encrypted. As stated in [20], minimum of 12.5% of the data, which is equivalent to one bitplane, needs to be encrypted, to distort the image. PSNR is again used as a metric to

¹ In JPEG compression standard [23], the bits that are representing the (AMPLITUDE) symbol followed by (RUNLENGTH, SIZE) symbol.

measure the level of encryption and classical encryption algorithm like AES (Advanced Encryption Standard) is used for encryption purposes. The proposed scheme in [20] is designed for uncompressed images. In [11], a real-time video encryption scheme is proposed in frequency domain. Where the first 10 DCT coefficients in 8x8 DCT block (DC and 9 AC's), are first XOR with the random numbers generated by Pseudo-Random Number Generator (PRNG). For DC Coefficients the PRNG generated values falls in 50 to 255, because DC coefficient has higher values and for AC coefficients the values fall in 0 to 30. After that in the next step a 1x64 vector is generated by PRNG to permute the DCT coefficients with in the macro block. In this paper the experiment showed that 86.53% of the DC values fall in 50-255 and 96.94% of the AC values fall in 0-30. A maximum of 23.41% of increase in the video size was introduced by the proposed encryption scheme and it encrypts one frame in average of 7.2 milliseconds. The proposed scheme encrypts the whole DC coefficient and also the perceptual quality has not been measured. In [21], it is showed that about 60% of the information is guaranteed by authenticating the DC coefficient and 80% of the information can be hindered by encrypting the DC coefficient. A set of experiments are performed in this paper to show that how much information the DC coefficient possesses. The paper states that the SSIM score was 0.7 when leaving the first AC coefficient and DC coefficient together and setting all the coefficients equal to zero. Another experiment was done by randomly flipping the sign bits of the DCT coefficients which result in SSIM value of 0.5.

III. THE PROPOSED TECHNIQUE

The Basic idea is to manipulate the energy contained by the DCT block. As reported in [11] and [22], a large amount of energy is present in the DC and the first few AC Coefficients. DC coefficient always has the largest amount of the energy among the rest of the DCT coefficients [24]. That is why we will split the DC coefficients in to bitplanes and gradually encrypt those bitplanes to get more levels for perceptual encryption. For the selection of the number of AC coefficient an experiment was carried out to see how much energy AC coefficients retains for that purpose a statistical analysis of the distribution of the AC coefficients of large number of images was performed. The normalized graph plotted in Fig. 1 shows the average values of the AC coefficients in the DCT block. A set of images were taken and their absolute AC coefficient values were summed. **It is clear from Fig. 1, that first 5 AC coefficient have large values.** And the most of the energy is concentrated in between the first 30 values of the DCT block. Logically, manipulating the higher AC coefficient values can give a good level of perceptual encryption. Also it is suggested in [22] that the encryption of the first few coefficients is sufficient to distort the content.

For our experiment, based on the observation that the first 5 AC coefficients have largest values in the DCT block, so the study is divided in to two cases in the first case first 5 AC coefficients were taken and are scrambled then DC

bitplanes were taken and scrambled along the 5 AC coefficients.

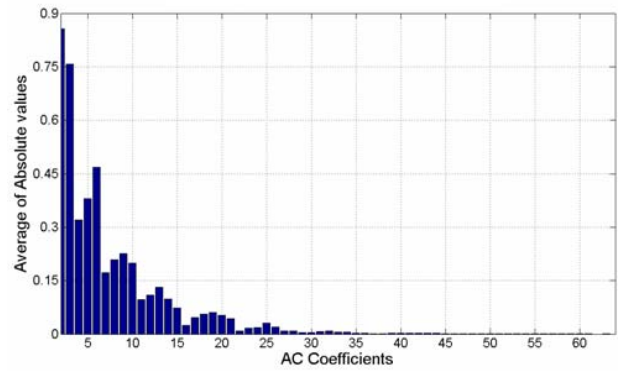


Figure 1. The graph shows the average normalized values of AC coefficients.

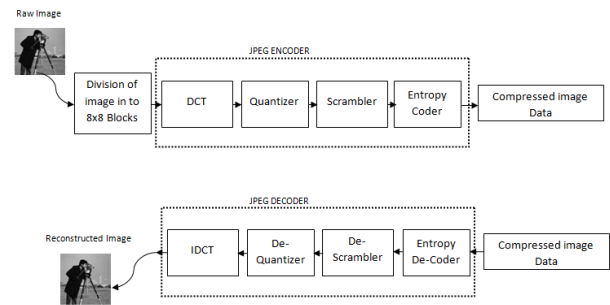


Figure 2. JPEG Coder and Encoder with Scrambler and Descrambler embedded in it respectively.

In order to scramble the quantized DCT coefficients a scrambler is placed after the quantization block in the JPEG image compression standard. The DCT coefficients are numbered according to zigzag order as shown in the Fig. 3 below.

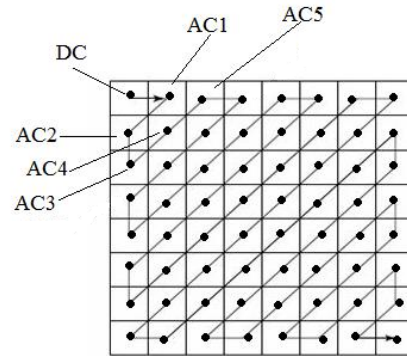


Figure 3. DCT Coefficient numbered according to Zig-Zag order.

IV. ENCRYPTER / SCRAMBLER

A Chaotic Seed Based Scrambler (SBR) was adopted to perform scrambling. This Chaotic Seed Based Scrambler is discussed in detail in [12]. The seed in SBR, act as initial condition or key which is used to generated the random positions. It's Cryptographic Security like key sensitivity analysis, spread and dispersion have also been discussed in [12]. The basic idea is usage of chaotic logistic map as given in (1),

$$x_n + 1 = rx_n (1 - x_n) \quad (1)$$

where $0 < x_n < 1$ and $3.47 < r < 4$, When the value of “ r ” is between 3.47 and 4, it gives random values. A scrambling matrix is generated by using chaotic logistic map in (1). The scrambling matrix has exactly one “1” in its each column or each row. The described scrambler has optimal spread and dispersion property. Spread measures the distance between the elements that were close before the permutation. And dispersion measures the randomness in spread. These two properties are important in our case as the coefficients and the bitplanes should be randomly scrambled and there spread and dispersion should be optimal to achieve quality encryption.

$$\begin{pmatrix} 0 & 0 & \dots & 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 1 & 0 & 0 \\ 0 & 0 & \dots & 0 & \dots & 0 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 0 & 0 & 0 & 1 \\ 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 1 & \dots & 0 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} X1 \\ X2 \\ X3 \\ X4 \\ \vdots \\ X1022 \\ X1023 \\ X1024 \end{pmatrix} = \begin{pmatrix} Y1 \\ Y2 \\ Y3 \\ Y4 \\ \vdots \\ Y1022 \\ Y1023 \\ Y1024 \end{pmatrix}$$

$S \cdot X = Y$

Here Y is the Scrambled output, S is the scrambling matrix generated by the chaotic logistic map and X is the input values.

V. METHOD OF PERFORMANCE EVALUATION

In past, there are not much metrics to measure the level of the perceptible distortion in the images after encryption process. Usually (Peak-Signal-to-Noise-Ratio) and MSE (Mean Square Error) [25] were used, but logically the metric which measure the level of perceptual security of the multimedia content should be based on the understanding of the image/video. In other words it should calculate the amount of the information is being perceived from the image or video. PSNR and MSE are used to calculate error introduced in the content. These measures are not suitable to measure the amount of information visible through the image/video [13]. There are some recently designed objective image quality assessment metrics like SSIM, MS-SSIM, VIF, VIFP, UQI and IFC, which are specially designed to measure the amount of perceived information. We will use these metrics to calculate the amount of perceived and hidden² (distorted) information in the image.

VI. RESULT AND ANALYSIS

First we will take first 5 AC coefficients of each quantized DCT block as they have higher values and then later 5 AC coefficients will be encrypted with DC bitplanes.

A. Using First 5 AC Coefficients

The first 5 AC coefficients are selected from the zig-zag order. The process is commenced by scrambling the only the 5th AC coefficient and then 5th and the 4th AC coefficients were scrambled together and then 5th, 4th and 3rd



Figure 4. Original Cameraman image. (SSIM = 1, MS-SSIM = 1, VIF = 1, VIFP = 1, UQI = 1).

AC coefficients were scrambled together. This goes on up to the 1st AC coefficient. So that entire first 5 AC coefficients were scrambled. After every step of AC coefficient scrambling, the values of SSIM, MS-SSIM, VIF, VIFP, UQI, IFC, VSNR (see [26] for more detail about VSNR) and PSNR were calculated and are shown in the Table 1.

Here it is noticed as we increase the number of AC coefficients in the scrambling process, the level of visible information through the content decreases. After encrypting the all 5 AC coefficients one can hide 43% to 64 %, (estimated in terms of SSIM) and 21% to 40% (estimated from MS-SSIM) of perceivable information in the image. On average 55.4% and 33.6% of the information is hidden estimated by SSIM and MS-SSIM, respectively. The percentage of the perceivable hidden information varies image to image. Here it is worth mentioning that the proposed scrambling scheme only scrambles the coefficients in its own orbit or in other words if AC 5 of any quantized DCT block is scrambled then that AC 5 can only take place of AC 5 in any other quantized DCT block, it cannot take place of AC 4 or AC 3 or any other DCT coefficient.



Figure 5. Cameraman image with its first five AC coefficients scrambled. (PSNR = 18.31, SSIM = 0.3878, MS-SSIM = 0.5902, VSNR = 7.47, VIF = 0.0927, VIFP = 0.0878, UQI = 0.2240)

B. Perceptual Encryption Using DC Bitplanes Along With AC Coefficients

As DC coefficient contains most of the energy in the DCT block [24], so one can manipulate the energy stored inside DC coefficients according to its need. The main idea here regarding the DC Coefficient in our experiment is to split

² Here ‘Hidden’ refers to the amount of the information that is distorted and is not clear or visible.

DC coefficient in 7 bits. After this division, all the bits corresponding to DC values of the DCT block were collected as shown in Fig. 6 and are arranged from least significant bits (LSB) to most significant bits (MSB). Each Bitplane starting from least significant bits (LSB) was scrambled, along with the 5 AC coefficients. In other words first the LSB was scrambled along with the 5 AC coefficients then 2nd to least significant bits and least significant bits along with the 5 AC coefficients are scrambled and so on until the most significant bit along with the all the DC bitplanes and 5 AC coefficients were scrambled. After each step of scrambling, SSIM, MS-SSIM, VIF, VIFP, UQI, IFC, VSNR and PSNR were calculated as shown in Table 1 and 2.

Mathematically 66 - 81% (estimated by SSIM value) and 66 - 77% (estimated by MS-SSIM value) of the perceivable information is now hidden. And on average 74.58% (SSIM)³ and 72.96% (MS-SSIM)⁴ of visual distortion is introduced. It can be seen that after scrambling all the DC bitplanes and 5 AC coefficients, the information through the image is not visible any more except for the edges as shown in the Fig 9. These edges can also be made unclear by increasing the number of AC Coefficients to be scrambled. An experiment was also performed with 9 AC coefficients, as they also contain high values and as used in [11]. It was noticed that 9 AC coefficients were sufficient to distort the edges which are visible in Fig. 9.

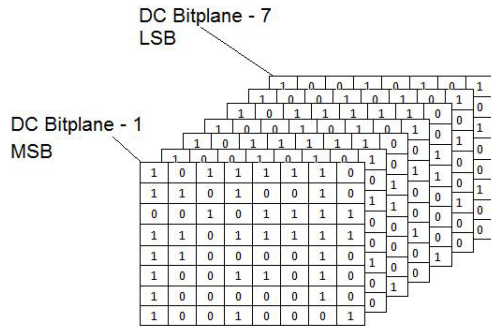


Figure 6. Collection of DC Coefficient bit planes



Figure 7. A progressive degradation is shown where all 5 AC Coefficients and 4 least significant bitplanes are scrambled together. (PSNR = 17.77, SSIM = 0.3617, MS-SSIM = 0.5561, VSNR = 6.82, VIF = 0.0844, VIFP = 0.0802, UQI = 0.1979)

³ Refers to the value estimated by SSIM.

⁴ Refers to the value estimated by MS-SSIM.

TABLE I
THE CORRESPONDING METRICS AVERAGE VALUES AGAINST EACH LEVEL OF PERCEPTUAL ENCRYPTION (FROM AC5 TO DC1)

Metric /Image	AC5	AC4	AC3	AC2	AC1	DC1
PSNR	29.4441	27.4349	26.0159	22.7847	20.8268	20.8162
SSIM	0.8169	0.7413	0.6727	0.5487	0.4460	0.4450
MS-SSIM	0.9426	0.9075	0.8705	0.7631	0.6640	0.6632
VSNR	22.0894	19.1006	17.0917	13.0001	10.6156	10.6130
VIF	0.4781	0.3487	0.2518	0.1655	0.1000	0.0997
VIFP	0.4198	0.3311	0.2784	0.1698	0.1062	0.1058
UQI	0.6408	0.5555	0.4915	0.3680	0.2719	0.2712
IFC	3.2307	2.1930	1.5199	0.9861	0.5959	0.5943

TABLE II
THE CORRESPONDING METRICS AVERAGE VALUES AGAINST EACH LEVEL OF PERCEPTUAL ENCRYPTION (FROM DC1 TO DC7)

Metric /Image	DC2	DC3	DC4	DC5	DC6	DC7
PSNR	20.7745	20.6188	20.0835	18.5233	15.6912	12.4689
SSIM	0.4421	0.4338	0.4126	0.3682	0.3060	0.2542
MS-SSIM	0.6600	0.6494	0.6191	0.5492	0.4391	0.2704
VSNR	10.5751	10.4171	9.8851	8.2722	5.2297	2.0637
VIF	0.0985	0.0959	0.0902	0.0809	0.0699	0.0597
VIFP	0.1048	0.1015	0.0940	0.0825	0.0684	0.0522
UQI	0.2691	0.2634	0.2497	0.2227	0.1833	0.1466
IFC	0.5882	0.5735	0.5415	0.4884	0.4248	0.3630

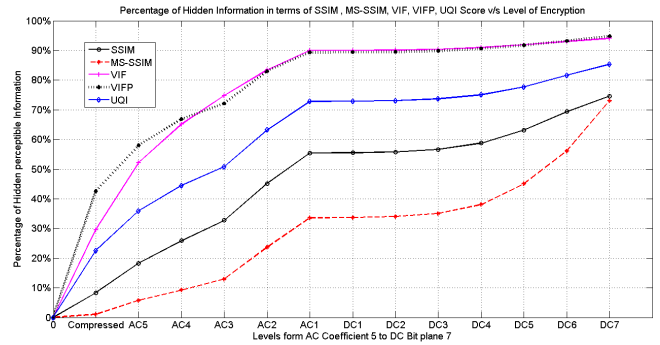


Figure 8. Graph showing the 5 AC coefficients and 7 DC Bitplanes v/s the percentage of hidden information measured in terms of SSIM, MS-SSIM, VIF, VIFP, and UQI.

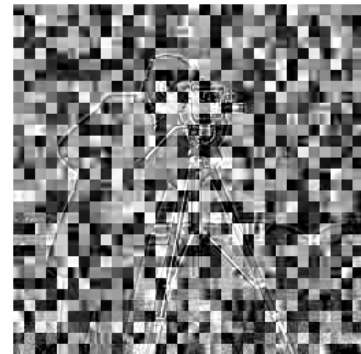


Figure 9. All 7 bitplanes of DC coefficients and 5 AC Coefficients scrambled together, of cameraman image. (PSNR = 8.49, SSIM = 0.1185, MS-SSIM = 0.0557, VSNR = -1.94, VIF = 0.0348, VIFP = 0.0170, UQI = 0.0713)

It is also noticed the values of metrics did not showed significant increase on the encryption of DC bitplanes as compared to values estimated by encrypting the DC bitplanes of small sized images. Also a certain level of

linearity is seen between the amount of distorted information and the encrypted number of coefficients and DC bitplanes. Especially in the case when only AC coefficients are encrypted.

C. Compression v/s Encryption

The RUNLENGTH coding in JPEG compression standard is based on 8x8 block. The scrambling is performed on the 8x8 DCT blocks and only the positions of those AC coefficients are changed, this affects the RUNLENGTHs. In case of DC coefficient, the differences of the DC coefficients are coded. In our case of DC bitplane scrambling, the values of DC coefficient will be changed after scrambling. The compression ratio is also calculated in the above mentioned experiment after the encrypting each AC Coefficient and DC bitplane.

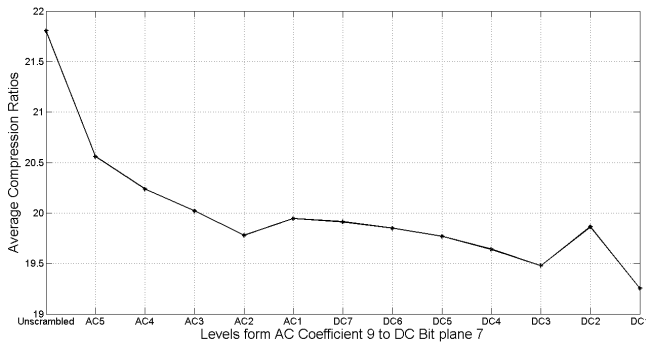


Figure 10. Average Compression Ratio against each level of Perceptual Encryption

In above mentioned cases of scrambling there are minor effects on the compression ratio. As also shown in the graph in Fig. 10, no significant change in compression ratio was observed after encryption process.

D. Security Analysis

There are many attacks that can be conducted on the encryption algorithm. Let us see the scenario in which only 9th AC coefficient is being scrambled. Consider the image is of size 256x256. For the JPEG processing it will be divided in to 1024 block of size 8x8. So the total possible combinations required to check for one coefficient will be 1024!. If 9 AC and 7 DC Bitplanes are scrambled all together then there are 64512! possible combinations. And exhaustive searches of combination are needed to be carried out. As the multimedia encryption scheme is considered to be secure if the cost of breaking is higher than the cost of buying that content and the time required to break the encryption algorithm is more than the time in which that content is considered to be useful [24].

E. Discussion

Recent research has shown that the recovery of DC coefficient is possible from AC coefficients as shown in [27] and [28], so we choose to encrypt AC as well as DC (in form of Bitplanes). It is possible that encrypted AC can be recovered from other unencrypted DCT coefficients. But we assume this is much more difficult since AC coefficients of adjacent blocks have less correlation than DC ones. In the future we will also study if and how the previous AC to DC recovery attacks may be extended to a more general

form. This will help us get a better understanding of security evaluation of DCT-based selective encryption.

The effects of the scrambling of the selected DCT coefficients which possess the most of the energy in the DCT block is investigated in the abovementioned experiments. A set of different images and of different sizes was taken and the corresponding average SSIM, MS-SSIM, VSNR, VIF, VIFP, UQI and IFC values are calculated and shown in the Tables 1 and 2 (MATLAB implementation of Image quality assessment can be found on [29]). To calculate all the values, the uncompressed original image was taken as a reference image. The first value in graphs in Fig. 8 shows the average measured value after compression.

VII. CONCLUSION

In this work a perceptual encryption scheme is proposed that used AC coefficients and DC bitplanes. The corresponding level of perceivable amount of information is measure with various metrics. The splitting of DC coefficient in to bitplanes will allows the design of the perceptual encryption scheme to have more control on the levels of encryption.

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